

Multiple integrals – 2

Change of variables and the Jacobian

Homework

13 сентября 2026 года

Theoretical minimum. The change of variables multiplies the integrand by $|\det D\varphi|$, and for a linear map this is the geometric meaning of the determinant. Three substitutions cover almost everything: polar, cylindrical and spherical, with $dx = r^{n-1} dr d\sigma(\omega)$ in $\mathbb{R}^n$; a radial integrand then becomes a one-dimensional integral against r^{n-1} . From polar coordinates one gets the Gaussian integral, the volume of the ball $\frac{\pi^{n/2}}{\Gamma(n/2+1)}$ and, after diagonalisation, the Gaussian integral with a quadratic form. Integrals over simplices and octahedra are the Dirichlet and Liouville formulas. Two further reflexes: a substitution that maps the domain onto itself proves identities rather than computing integrals, and a bad Jacobian at the boundary is harmless as long as the exceptional set has measure zero.

1. (VJIMC 2015) Find all continuously differentiable $f: \mathbb{R} \rightarrow \mathbb{R}$ such that for every $a \geq 0$

$$\iiint_{D(a)} x f\left(\frac{ay}{\sqrt{x^2 + y^2}}\right) dx dy dz = \frac{\pi a^3}{8} (f(a) + \sin a - 1),$$

where $D(a) = \{(x, y, z) : x^2 + y^2 + z^2 \leq a^2, |y| \leq x\sqrt{3}\}$.

2. Compute $\iint_{\mathbb{R}^2} e^{-(x^2+xy+y^2)} dx dy$ and, more generally, $\int_{\mathbb{R}^n} e^{-\langle Ax, x \rangle + \langle b, x \rangle} dx$ for a positive definite A .
3. Compute the volume of the ellipsoid $\sum \frac{x_i^2}{a_i^2} \leq 1$, and the volume of the intersection of the unit ball with the half-space $x_1 \geq h$ in $\mathbb{R}^n$.
4. Using the substitution $x = uv, y = u(1-v)$, compute $\int_{x,y \geq 0, x+y \leq 1} x^{p-1} y^{q-1} dx dy$ and deduce the relation between the beta and the gamma functions (Integrals – 5).

5. Compute $\iiint_{x^2+y^2+z^2 \leq 1} \frac{dx dy dz}{\sqrt{x^2+y^2+(z-a)^2}}$ for $a > 1$ and for $0 < a < 1$, and interpret the answer as the Newtonian potential of a ball.
6. Prove that $\int_{\mathbb{R}^n} f(\|x\|) dx = \sigma(S^{n-1}) \int_0^\infty f(r) r^{n-1} dr$ and use it to decide for which p the integrals $\int_{\|x\| \leq 1} \|x\|^{-p} dx$ and $\int_{\|x\| \geq 1} \|x\|^{-p} dx$ converge.
7. Prove that the Jacobian of the inverse map is the reciprocal of the Jacobian, and compute the area of the region bounded by the curves $xy = 1$, $xy = 2$, $y = x$, $y = 4x$ by a suitable substitution.
8. Compute $\iint_{[0,1]^2} \frac{dx dy}{(1+x^2+y^2)^2}$ and $\iint_{x^2+y^2 \leq 1} \frac{dx dy}{\sqrt{1-x^2-y^2}}$.
9. Prove that a substitution mapping the disc onto itself and preserving the area gives the identity $\iint_D f(\varphi(x, y)) dx dy = \iint_D f$, and use a rotation to evaluate $\iint_{x^2+y^2 \leq 1} (ax + by)^2 dx dy$.
10. Compute the n -dimensional volume of the set $\{x \in \mathbb{R}^n : |x_1| + \cdots + |x_n| \leq 1\}$ and of $\{x : \max |x_i| \leq 1\}$, and find the ratio of the volumes of the ball and of the cube as $n \rightarrow \infty$.